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Successive multivariate variational mode decomposition

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extracting mode has no or less spectral overlap with the previously obtained modes and the residual signal. Finally, we employ the alternate direction method of the multiplier algorithm (ADMM) to solve it. Compared with other multivariate extending methods whose performances will be degraded if the number of modes is not precisely known, this extension can recursively extract modes and does not need to know the number of modes. Therefore, it achieves better performance on convergence and computation requirements. Moreover, it is more robust to the initial center frequency and possesses the mode-alignment property. We also investigate the relationships between the regularization parameter α and the spectrum property of modes. Some suggestions for selecting proper solution parameters are provided. Finally, we show promising practical decomposition results on a series of simulating and real-life multi-channel data.

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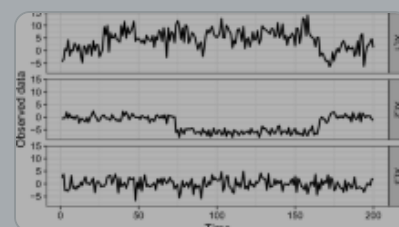
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```
8:  $f(\omega, \tau_q) = f(\omega, \tau) \otimes \delta(\tau - \tau_q)$ 
9: (2) Initialize the iteration counter, center frequency
10: If  $q = 1$  then
11:    $\omega_q^0(\tau_q) = \arg \max_{\omega} f(\omega, \tau_q)$ 
12: else if  $q > 1$  then
13:    $\omega_q^0(\tau_q) = \omega_{q-1}^0(\tau_{q-1})$ 
14: end if
15:  $n \leftarrow 1$ 
16: (3) Extract windowed mode
17: repeat
18:   Update  $\hat{\omega}_n$ :
19:   
$$\hat{\omega}_n^{k+1}(\omega, \tau_q) = \frac{\int_{\tau_q} f(\omega, \tau_q) - \alpha^2 (\omega - \omega_q^k(\tau_q))^2 \hat{\omega}_n^k(\omega, \tau_q) + \frac{1}{2} \log(\tau_q)}{(1 + \alpha^2 (\omega - \omega_q^k(\tau_q))^2) (1 + 2\alpha (\omega - \omega_q^k(\tau_q))^2)}$$

20:   Update  $\hat{\omega}_n$ :  $\omega_q^{n+1}(\tau_q) = \frac{\int_{\tau_q} \omega |\hat{\omega}_n^{k+1}(\omega, \tau_q)|^2 d\omega}{\int_{\tau_q} |\hat{\omega}_n^{k+1}(\omega, \tau_q)|^2 d\omega}$ 
```



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References

Bao, C., Hao, H., Li, Z. X., & Zhu, X. (2009). Time-varying system identification using a newly improved HHT algorithm. *Computers & Structures*, 87(23), 1611–1623.

[Article](#) [Google Scholar](#)

Chen, Q., Xie, L., & Su, H. (2020). Multivariate nonlinear chirp mode decomposition. *Signal Processing*, 176, 107667.

[Article](#) [Google Scholar](#)

Chen, S., Dong, X., Peng, Z., Zhang, W., & Meng, G. (2017). Nonlinear chirp mode decomposition: A variational method. *IEEE Transactions on Signal Processing*, 65(22), 6024–

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Feng, Z., Liang, M., & Chu, F. (2013). Recent advances in time–frequency analysis methods for machinery fault diagnosis: A review with application examples. *Mechanical Systems and Signal Processing*, 38(1), 165–205.

[Article](#) [Google Scholar](#)

Hao, H., Wang, H. L., & Rehman, N. U. (2017). A joint framework for multivariate signal denoising using multivariate empirical mode decomposition. *Signal Processing*, 135, 263–273.

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Jiang, X., Shen, C., Shi, J., & Zhu, Z. (2018). Initial center frequency-guided VMD for fault diagnosis of rotating machines. *Journal of Sound and Vibration*, 435, 36–55.

[Article](#) [Google Scholar](#)

Jiang, X., Wang, J., Shi, J., Shen, C., Huang, W., & Zhu, Z. (2019). A coarse-to-fine decomposing strategy of VMD for extraction of weak repetitive transients in fault diagnosis of rotating machines. *Mechanical Systems and Signal Processing*, 116, 668–692.

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Li, Z., Jiang, Y., Guo, Q., Hu, C., & Peng, Z. (2018). Multi-dimensional variational mode decomposition for bearing-crack detection in wind turbines with large driving-speed variations. *Renewable Energy*, 116, 55–73.

[Article](#) [Google Scholar](#)

Liu, S., & Yu, K. (2022). Successive multivariate variational mode decomposition based on instantaneous linear mixing model. *Signal Processing*, 190, 108311.

[Article](#) [Google Scholar](#)

Looney, D., & Mandic, D. P. (2009). Multiscale image fusion using complex extensions of emd. *IEEE Transactions on Signal Processing*, 57(4), 1626–1630.

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Rehman, N., & Mandic, D. P. (2010). Multivariate empirical mode decomposition. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 466(2117), 1291–1302.

[Article](#) [MathSciNet](#) [Google Scholar](#)

Rehman, N., Looney, D., Rutkowski, T.M., & Mandic, D.P. (2009). Bivariate EMD-based image fusion. In: 2009 IEEE/SP 15th Workshop on Statistical Signal Processing, pp. 57–60

Rehman, N., Xia, Y., & Mandic, D.P. (2010). Application of multivariate empirical mode decomposition for seizure detection in EEG signals. In: 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology, pp. 1650–1653

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Sun, H., Fang, L., & Zhao, F. (2019). A fault feature extraction method for single-channel signal of rotary machinery based on VMD and KICA. *Journal of Vibroengineering*, 21(2), 370–383.

Wu, Z., & Huang, N. E. (2009). Ensemble empirical mode decomposition: A noise-assisted data analysis method. *Advances in Adaptive Data Analysis*, 01(01), 1–41.

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Appendix A

In this appendix, we show that Eq. (28) is tenable around the center frequency ω_k and has the same effective bandwidth as Eq. (6). The filter in Eq. (20) can be expressed as:

$$H_k(\omega, \omega_k, \left\{ \omega_i \right\}_{i=1}^{k-1}) = \frac{1}{1 + 2\alpha \left(\omega - \omega_k \right)^2 + \sum \limits_{i=1}^{k-1} \left\{ \frac{1}{\alpha^2} \left(\omega - \omega_i \right)^4 \right\}}$$

(A.1)

Assume that the algorithm of SMVMD is converged to the true modes and also, the modes are spectrally distinct. When ω is close to ω_k , then $\left| \omega - \omega_i \right| \geq \rho > 0$ and the following relation is given:

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(B.1)

where $(\{\{\{\vec{a}\}_k\})$ with the same size as $(\{\{\{\hat{u}\}\}_k(\omega)\})$ is the k -th receiver steering vector and $(\{\{\{\hat{s}\}\}_k(\omega)\})$ denotes the corresponding source. The following operation can be conducted:

$$\begin{aligned} & \int_{-\infty}^{+\infty} \{\{\{\vec{a}\}_k(\omega)\} \cdot \{\{\{\hat{u}\}\}_{k,c}^{H}(\omega)\} \mathrm{d}\omega \\ & = \int_{-\infty}^{+\infty} \{\{\{\vec{a}\}_k\} \{\{\{\hat{s}\}\}_k(\omega)\} \cdot \{a_{k,c}\} \{\{\{\hat{s}\}\}_k^H(\omega)\} \mathrm{d}\omega \\ & = \{\{\{\vec{a}\}_k\} \int_{-\infty}^{+\infty} \{a_{k,c}\} \left| \{\{\{\hat{s}\}\}_k(\omega) \right|^2 \mathrm{d}\omega \end{aligned}$$

(B.2)

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$$\int_{-\infty}^{+\infty} \{\hat{u}\}_{k,1}(\omega) \cdot \{\hat{u}\}_{k,c}^H(\omega) \, \mathrm{d} \omega \quad (B.4)$$

Then $\{\hat{s}\}_k(\omega)$ can be recovered based on Eq. (B.1):

$$\begin{aligned} \frac{\{\vec{a}\}_k^H \{\hat{\vec{u}}\}_k(\omega)}{\{\vec{a}\}_k^H \{\vec{a}\}_k} \end{aligned} \quad (B.5)$$

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